

EE4C10

Analog Circuit Design

Fundamentals

Week 2

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Today's goal

- Recap basic concepts and the fundamentals of CS stage with quizzes
- Go through 2 LT spice exercises

Acknowledgments

Some images from B.Razavi's "Design of analog CMOS integrated circuits", copyright of McGraw-Hill.

Q1

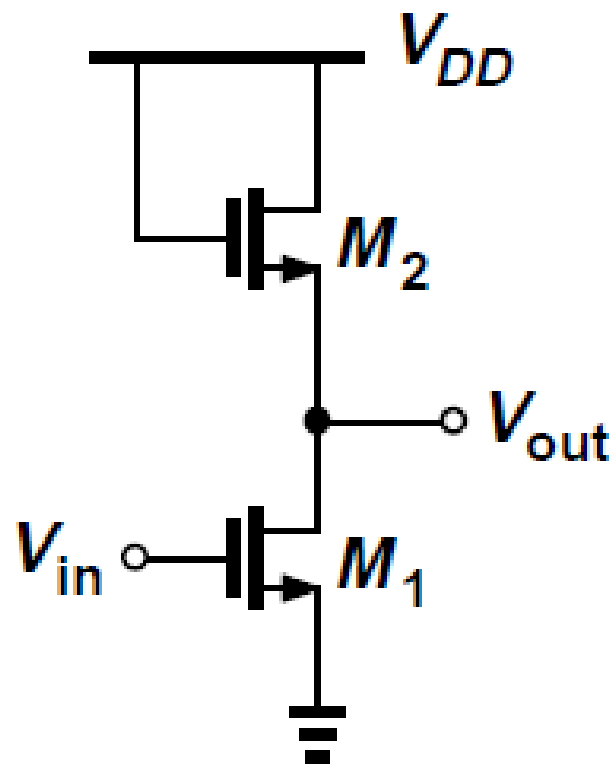
- What is the desired input and output impedance of a current-voltage amplifier
 - A. High input impedance; high output impedance
 - B. High input impedance; low output impedance
 - C. Low input impedance; high output impedance
 - D. Low input impedance; low output impedance

Q2

- What kind of simulation can be used to check the bias condition of an amplifier
 - A. AC simulation
 - B. DC simulation
 - C. Transient simulation
 - D. None of above

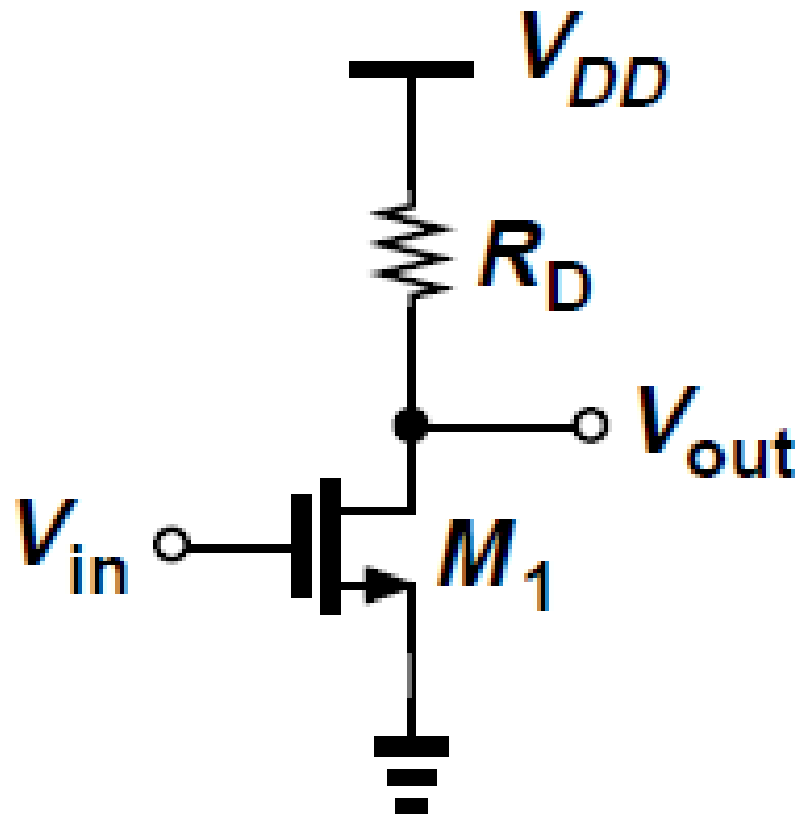
Q3

- What is the output voltage swing of this amplifier, to keep both transistors in saturation?
 - A. $V_{dsat1} \sim (V_{DD} - V_{th2})$
 - B. $0 \sim V_{dsat2}$
 - C. $V_{dsat1} \sim (V_{DD} - V_{dsat2})$
 - D. $0 \sim V_{DD}$



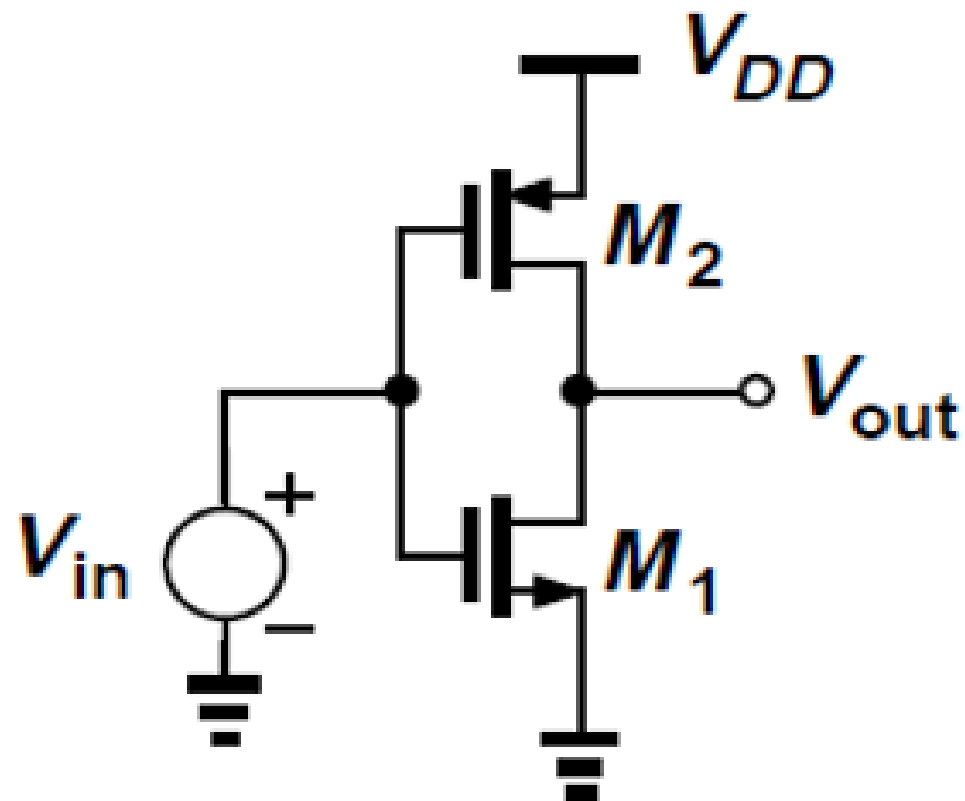
Q4

- What would you do to increase the gain of this amplifier, assume R_D is much smaller than r_{o1} ?
 - A. Replace R_D by a current source
 - B. Decrease the transconductance of M_1
 - C. None of above



Q5

- What are the advantages of inverter-based amplifier?
 - A. High gain
 - B. Low supply sensitivity
 - C. Low PVT variation



LTSpice Simulation Session

- We have 2 exercises.
- Download the exercises from bright space:
- LTspice files for practicals=> Week 2=> Week2 Exc1, Week2 Exc2

Exercise 1a

- Open Exc1.asc, you will see two CS stage amplifiers. Amplifier A has a current source load and amplifier B has a resistor load.
- Do a DC operating point simulation to find out whether M_1 and M_2 are biased in saturation.
- Calculate the DC gain of both amplifiers.
- Do an AC simulation to plot the gain of the two amplifiers. Does the simulation agree with your calculation?

Exercise 1b

- Now change I_{D1} so that the drain current of M_1 equals to that of M_2
- Do a DC operating point simulation. Are both transistors in saturation?
- Calculate the DC gain of both amplifiers
- Do an AC simulation to plot the gain of the two amplifiers. Does the simulation agree with your calculation?

Exercise 2

- Open Exc2.asc, you will see a resistive-degenerated CS stage with current source load. The current source is implemented by a PMOS transistor: M_{pc2} .
- Here we have a simple “current mirror: M_{pc1} and M_{pc2} . Since their gates are connected together, they have the same V_{gs} . And since they have also the same dimension, they have “**almost**” the same drain current. So we say M_{pc2} “mirrors” the current from M_{pc1} .
- Do a DC operating point simulation to find out whether all the transistors are in saturation.

Exercise 2

- Calculate the absolute DC gain of the amplifier
- Do an AC simulation to plot the gain and compare it with your calculation.